YZV 322E — APPLIED DATA ENGINEERING

DBeaver

Universal Database Manager for Data Engineers

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WHAT IS IT?

A free, open-source, universal database client

One single application to connect to, query, and manage any database — PostgreSQL, MySQL, SQLite, MongoDB, and 80+ more.

First Release	2010
Maintainer	DBeaver Corporation
License	Apache License 2.0 (Free)
Editions	Community (Free) & PRO (Paid)

WHAT PROBLEM DOES IT SOLVE?

One tool instead of many

Data engineers work with multiple database systems. Each one typically requires a separate client tool. DBeaver replaces them all with a single, unified interface.

- PostgreSQL
- MySQL
- SQLite
- MongoDB
- Oracle
- SQL Server
- ClickHouse
- + 80 more

HOW DOES IT COMPARE?

DBeaver vs. Alternatives

vs. pgAdmin

pgAdmin only supports PostgreSQL. If your team switches databases, you need a new tool. DBeaver supports all of them from the same interface.

vs. Command Line (psql)

The terminal is powerful but has no visual feedback — no diagrams, no click-to-filter, and a steep learning curve for beginners.

DBeaver Advantage

- Universal — works with 80+ databases
- Visual ER diagrams (auto-generated)
- Built-in JSON viewer
- Visual execution plan
- Drag-and-drop query builder
- Data export (CSV, JSON, Excel)

COURSE CONNECTION

Where DBeaver fits in the pipeline

DBeaver relates to the PostgreSQL & pgAdmin weeks of our course.
It sits at the end of the data pipeline as the observation and debugging layer.

Docker → Airflow → PostgreSQL → **DBeaver**

*"While Airflow moves the data and Docker runs everything,
DBeaver is where we understand the data."*

SETUP & INSTALLATION

Docker-based, fully reproducible

Clone the repo and start the database:

```
# Clone the repository
git clone <repo-url>

# Start the PostgreSQL database
docker compose up -d

# Install DBeaver (macOS)
brew install --cask dbeaver-community
```

What Docker sets up automatically:

- PostgreSQL 15 container
- 6 tables with foreign keys
- 4 indexes (including GIN for JSON)
- 2 database views
- 200+ sample records
- JSON metadata in reviews table

Live Demo

Switching to DBeaver UI

ER Diagram

Auto-generated schema visualization

Data Grid

Sort, filter, and edit data in-place

SQL Editor

Autocomplete, window functions, CTEs

Execution Plan

Visual query performance analysis

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JSON Viewer

Browse and edit JSONB data

Transactions

Manual commit and rollback safety

DEMO HIGHLIGHTS

Key features shown in the live demo

- **ER Diagram —** auto-generated from foreign keys, zero configuration
- **Data Grid —** sort, filter, and edit cells directly like a spreadsheet
- **SQL Autocomplete —** suggests table names, column names as you type
- **Execution Plan —** visual tree showing index usage and query cost
- **JSON Viewer —** formatted panel for JSONB columns with nested data
- **Manual Commit —** rollback mistakes before they hit the database

Limitation: Some features like Schema Compare and Data Compare are only available in the paid PRO edition. The Community edition covers all the essentials shown today.

REPOSITORY OVERVIEW

GitHub project structure

```
dbeaver-tool-presentation/
├── docker-compose.yml    # PostgreSQL setup
├── scripts/
│   └── seed_data.sql      # Schema + sample data
├── sample_queries.sql   # Demo queries
├── README.md           # Full instructions
├── ai_usage.md         # AI disclosure
└── slides.pdf         # This presentation
```

Anyone can clone, run **docker compose up -d**,
and reproduce the entire demo in under 2 minutes.

REFERENCES

Sources & Links

dbeaver.io/docs

Official DBeaver Documentation

github.com/dbeaver/dbeaver

DBeaver Open Source Repository

postgresql.org — JSON Functions

PostgreSQL 15 JSONB Documentation

postgresql.org — Window Functions

PostgreSQL 15 Window Functions Tutorial

hub.docker.com — postgres

Official PostgreSQL Docker Image

Thank you!